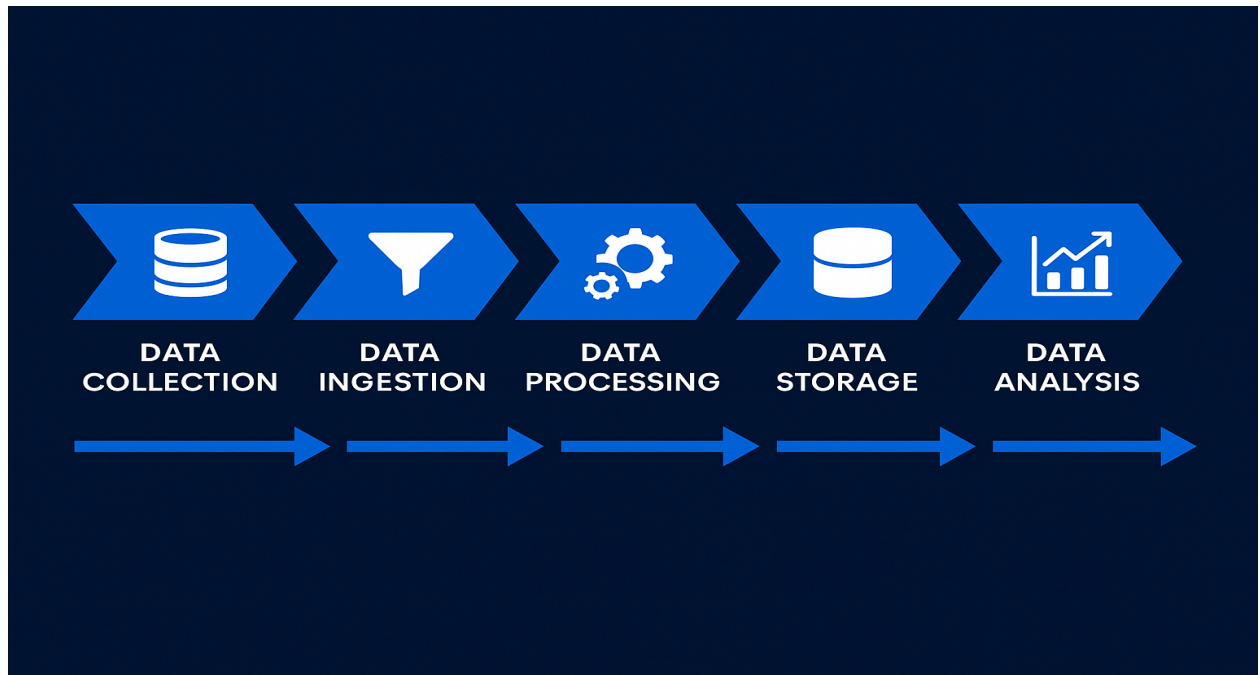




Stage 8 Pipeline Task

RetailCo Business Insight

Prepared by: RetailCo's Data Engineering Team
Group E (Chung Yok, Marvest, Yvonne Akoth, Felixanna)
2/6/2026

Introduction

RetailCo is a retail chain operating in Nigeria with four stores across Lagos, Abuja, Port Harcourt and Kano. For years the company has been running a legacy ERP system and now the company aims to modernize its data infrastructure through hiring a team of highly qualified data engineers. Our task as the data engineering team hired by RetailCo is to build a complete data platform that will strengthen all future reporting and analytics for the business, shutting down all data areas exposed through a REST API among other duties.

Response to the Five Business Questions Asked by Management Weekly

1. Revenue Performance: RetailCo generated total revenue of **63,059,212,667.78** during the analysed period, covering sales from June 2024 to June 2026.

Revenue was well distributed across the four stores, showing that the business is not overly dependent on one location. RetailCo Port Harcourt was the highest-performing store, generating 15,984,040,682.89 from 19,241 orders. This was followed by RetailCo Lagos with 15,768,107,008.41, RetailCo Kano with 15,718,041,019.48, and RetailCo Abuja with 15,589,023,957.00.

At category level, Books & Media was the strongest revenue driver, generating 9,445,511,122.62 from 271,360 units sold. It was followed by Food & Beverage with 8,833,067,033.13, and Clothing with 8,717,142,141.58. The lowest-performing category in the output was Health & Beauty, with 6,903,522,641.37 revenue and 208,271 units sold. Product-level results showed that key revenue drivers included Small Frozen Chips, Fantastic Concrete Keyboard, and Electronic Cotton Soap.

The monthly revenue trend showed consistently strong performance, with visible monthly revenue generally remaining in the multi-billion range. The highest visible month was July 2024, generating 2,784,678,344.99 from 3,274 orders. Overall, RetailCo's revenue performance appears stable across time, with strong contributions from all stores and major product categories.

2. Customer Behaviour: Customer behaviour was analysed through purchase frequency, average order value(AOV), and spending-based customer segmentation. The highest-frequency customer was Flavio Toy, who placed 30 orders. Others, including Aglae Waters, Ron Moen, Lavern Beer, Paul Beer, and Maximillia Osinski, placed 29 orders each. This shows that RetailCo has a group of repeat customers who purchase frequently and may be valuable for loyalty and retention campaigns. The overall AOV was 828,549.07. This means that each valid order contributed approximately 828.5k in revenue on average. This relatively high AOV suggests that customers may be purchasing multiple items per order, higher-value products, or both. Customer segmentation showed that revenue is strongly concentrated in the High Value segment. This segment contained 3,671 customers and generated 52,411,597,800.57 in revenue. The Medium Value segment contained 1,265 customers and generated 10,384,498,334.90, while the Low Value segment contained 64 customers and generated 263,116,532.31. These results show that high-value customers are the main driver of RetailCo's revenue. Management should focus on retaining these customers while encouraging medium-value customers to increase purchase frequency and basket size.

3. Product and Discount Analysis: RetailCo's best-selling products are spread across several categories. By units sold, Rustic Fresh Shoes was the top product, with 2,962 units sold and

98,027,059.81 in revenue. This was followed by Fantastic Concrete Keyboard, with 2,788 units sold and 113,049,663.66 in revenue. At category level, Books & Media sold the highest number of units, with 258,248 units sold and 8,988,863,634.17 in revenue. It was followed by Clothing with 249,827 units, Home & Garden with 242,616 units, and Food & Beverage with 238,412 units. Discounting was moderate across categories. Sports had the highest average discount rate at 3.60%, followed by Clothing, Food & Beverage, and Automotive, each around 3.51%. At product level, the highest average discounts were found in products such as Intelligent Metal Keyboard at 5.70%, Refined Soft Chair at 5.66%, and Gorgeous Plastic Computer at 5.55%.

Margin analysis showed that Books & Media produced the highest estimated margin at 2,260,844,224.17, followed by Food & Beverage with 2,155,625,188.69, and Sports with 2,063,449,753.74. Although Health & Beauty had lower total sales, it had the highest margin percentage at 26.15%, suggesting it may be more profitable per unit of revenue. RetailCo should prioritise high-volume and high-margin categories while monitoring heavily discounted products to ensure discounts are not reducing profitability unnecessarily.

4. Payment Channel Insights: RetailCo uses five payment methods: Cash, Bank Transfer, USSD, Mobile Money, and Card. By transaction count, Cash was the most used method, with 14,510 payment events and 11,202,810,041.92 in total amount paid. However, by total payment value, Bank Transfer ranked first with 11,321,526,979.32, followed by USSD with 11,299,625,548.42. Digital payments dominate the payment ecosystem. Digital methods accounted for 57,570 payment events and 44,798,565,843.73 in total amount paid, compared with 14,510 non-digital Cash payments worth 11,202,810,041.92. Although all 72,080 payment records were marked as completed, payment-quality checks identified 1,495 refunds, 765 negative non-refund payments, and 719 zero-amount payments. These records should be flagged for review because they may affect reconciliation and financial reporting.

5. Operational Data Quality: The operational data quality checks showed that the core warehouse data is largely reliable. The main table volumes were 359,221 order items, 80,000 orders, 72,080 payments, 5,000 customers, 2,000 products, 5 payment methods, and 4 stores. Duplicate primary ID checks returned no duplicate records, meaning the pipeline preserved uniqueness across key tables. Missing field checks also passed. Across 80,000 orders, there were no missing order IDs, customer IDs, store IDs, statuses, ordered dates, or total amounts. All 80,000 orders had valid total amounts, and all 359,221 order items had valid item values. Relationship checks also returned zero broken references between orders, order items, products, customers, stores, payments, and payment methods. This means the pipeline maintained referential consistency across the warehouse.

The main data quality concern was found in the payments table. Although all payment records were marked as completed, the pipeline identified refunds, zero-amount payments, and negative non-refund payments. These were handled through rule-based flags such as Normal payment, Refund payment, Negative non-refund payment, and Zero amount payment. This approach keeps the raw data auditable while allowing anomalous records to be reviewed or excluded from sensitive financial reporting.

Overall, the warehouse successfully answers the five management questions. Revenue is strong and balanced across stores, high-value customers drive most sales, Books & Media and related high-performing categories lead product performance, digital payments dominate the payment mix, and the core operational data is reliable. The main recommendation is to strengthen payment

validation and monitoring so that zero-value and negative non-refund payments are flagged early in the pipeline before management reporting.